

## Research Interests

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My research lies at the intersection of computer systems and AI. I study how emerging AI workloads challenge existing system abstractions by introducing new computational behaviors and application requirements. Based on these insights, I design efficient runtimes and resource-management mechanisms that adapt AI execution to heterogeneous computing platforms, from datacenter infrastructure to resource-constrained edge devices.

## Education

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### Korea Advanced Institute of Science and Technology (KAIST)

Sep. 2025 – Expected Aug. 2027

*Master of Science in Computer Science, School of Computing*

*Daejeon, Republic of Korea*

- Overall GPA 4.2/4.3; Recipient of the National Full Graduate Scholarship (Fall 2025 – Spring 2027)
- Advisor: Prof. Jaehyuk Huh, Computer Architecture and Systems Lab (CASYS)

### Sungkyunkwan University

Mar. 2020 – Aug. 2025

*Bachelor of Science in Computer Science and Engineering*

*Suwon, Republic of Korea*

- Overall GPA 4.2/4.5; Recipient of the Sungkyun Software and Research Promotion Scholarships (full tuition, 8 semesters)

## Publications

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### Beyond LLM Serving: Characterizing Vision-Language-Action Workloads for Embodied AI System Design

- **Seonghun Jung**, Sieun Moon, Jiyoung Jeong, Jimin Lee, and Jaehyuk Huh [ **Preprint** ]
- ASPLOS 2027

## Experience

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### Samsung Electronics | AI Center

Jan. 2025 – Feb. 2025

*Software Engineer Intern*

*Hwaseong, Republic of Korea*

- Reduced page-load latency by 26.45% on a semiconductor equipment data-analysis platform by tracing bottlenecks with Elastic APM/RUM and the ELK stack.
- Reduced server processing time by 6.49% in personalized dashboards by implementing caching and optimizing multi-table joins.

### Samsung Electronics | AI Center

Sep. 2024 – Dec. 2024

*MLOps Engineer Intern*

*Hwaseong, Republic of Korea*

- Developed a multi-tenant FastAPI backend for a distributed Ray/Kubeflow hyperparameter-optimization platform used by 100+ Samsung engineers, integrating Keycloak JWT authentication, Redis job tracking, and Volcano GPU scheduling.
- Automated platform deployment with ArgoCD, Helm, and Istio; centralized logging with FluentBit and OpenSearch.

### Computer System Lab | Prof. Euseong Seo

Sep. 2023 – Jun. 2024

*Undergraduate Researcher*

*Suwon, Republic of Korea*

- Designed a throughput-gradient scheduler that reallocates GPUs across distributed deep-learning jobs at runtime to improve cluster utilization.
- Built a 4-node Kubernetes/InfiniBand testbed with Containerd and the Kubeflow MPI Operator to run Elastic Horovod workloads.

## Projects

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### ProfilingSense

Sep. 2024 – May 2025

*Non-Intrusive Kubernetes Scheduler for Distributed DL Based on Resource Efficiency Profiling*

*B.S. Graduation Project*

- Designed and implemented a non-intrusive Kubernetes gang scheduler that ranks jobs using throughput-per-resource profiles and wait-time aging, without training-code changes or GPU reallocation.
- Reduced peak queue length by 8.2% and P99 JCT by 5.8% versus FIFO-Gang scheduling in an 864-GPU replay of 2,312 Microsoft Philly jobs using five workload profiles measured on a 6-GPU testbed.

## Honors & Skills

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**Honors:** Outstanding Teaching Assistant Award, KAIST (2025, 2026); Gold Prize (2nd Place), SKKU Hackathon (2023); 15th Place, SKKU Programming Competition (2024, 2025)

**Certifications:** Certified Kubernetes Administrator, The Linux Foundation (2025); S/W Certification Professional, Samsung Electronics DX (2024)

**Technical:** Python, C/C++, CUDA, PyTorch, Kubernetes, Docker, Linux, Nsight Compute/Systems (NCU/NSYS)